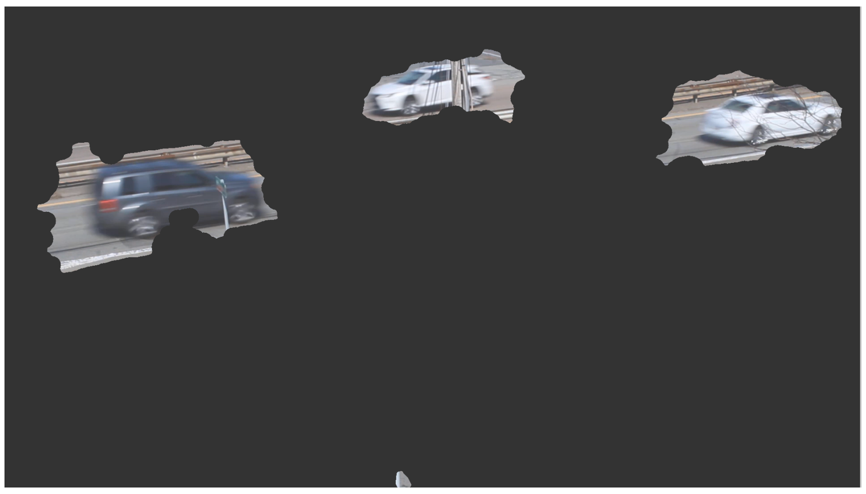
Project: Applying Optical Flow to Determine the Direction of Traffic

My Solutions >

You previously developed code to apply optical flow to a few frames from a video of a highway and created a mask of the fastest-moving objects:



Here, you will use optical flow and this mask to determine how many cars are traveling in each direction and their velocity.

In this problem, you will:

- Copy your code from the previous problem to calculate the optical flow and create a flow magnitude mask.
- Calculate the average v_x for each object in the mask.
- Store the average v_x for every object moving **to the left** *and* has an **average** v_x **above 3** in a variable named `leftVx`. These are your cars driving left.
- Store the average v_x for every object moving **to the right** *and* has an **average** v_x **above 3** in a variable named rightVx. These are your cars driving right.
- Store the total number of cars driving to the left as `numCarsLeft`.
- Store the total number of cars driving to the right as `numCarsRight`.

You will be graded based on the following:

- Correctly detecting the number of left-moving vehicles, `numCarsLeft`
- Correctly detecting the number of right-moving vehicles, `numCarsRight`
- Correct values for the array of left-moving velocities, `leftVx`
- Correct values for the array of left-moving velocities, `rightVx`

Script

Save   Reset   MATLAB Documentation (https://www.mathworks.com/help/)

```
frame1 = imread("Rt9Frame1.png");
frame2 = imread("Rt9Frame2.png");

% Paste your solution code from "Project: Applying Optical Flow to Detect Moving Objects" here:
of = opticalFlowFarneback;
estimateFlow(of,im2gray(frame1));
flow = estimateFlow(of,im2gray(frame2));
vm = flow.Magnitude;
mask = vm(:,:) > 1;
mask = bwareafilt(mask, [500 inf]);
mask = imclose(mask, strel('disk', 20, 0));


% Write new code to count the number of cars moving in each direction and their x-direction velocities
CC = bwconncomp(mask);
idxlists = CC.PixelIdxList;
leftVx = [];
rightVx = [];
numCarsLeft = 0;
numCarsRight = 0;
for i=1:CC.NumObjects
    component_idx = idxlists{i};
    mVx = mean(flow.Vx(ind2sub(size(mask), component_idx)));
    if abs(mVx) > 3
        if mVx < 0
           leftVx = [leftVx mVx];
           numCarsLeft = numCarsLeft + 1;
        else
           rightVx = [rightVx mVx];
           numCarsRight = numCarsRight + 1;
        end
    end
end
```

Run Script

Assessment: All Tests Passed

Submit

Correctly Count Left-Moving Cars

Correctly Count Right-Moving Cars

leftVx Has Correct Values

rightVx Has Correct Values

Output

```
Code ran without output.
```